Perturbed Coin Model Analysis

This notebook validates the symmetry-analysis infrastructure used in the perturbed coin model workflow.
It checks that the central classes and methods for tensor-product symmetry generators are available, including both the standard and SO(N)-specific implementations.
It also verifies method signatures and wrapper consistency in `MatrixProductState`, ensuring coefficient extraction via `find_generators_coefficients` is correctly exposed for downstream analysis.

```
======================================================================
MODULE VALIDATION CHECK
======================================================================

[1] Checking Centraliser.TensorProductGroup...
    ✓ TensorProductGroup class exists with required methods

[2] Checking Centraliser.TensorProductGroupSO (SO(N) framework)...
    ✓ TensorProductGroupSO class exists with required methods

[3] Checking Centraliser.SON (SO(N) basis)...
    ✓ SON class exists with so_basis method

[4] Checking find_generators_coefficients parameters...
    ✓ Parameters: ['self', 'M', 'verbose', 'basis_generators', 'reference_coeffs']

[5] Checking MatrixProductState.find_generators_coefficients wrapper...
    ✓ Wrapper method exists with parameters: ['self', 'M', 'basis', 'verbose', 'dim',
'reference_coeffs']
```

Form of the model

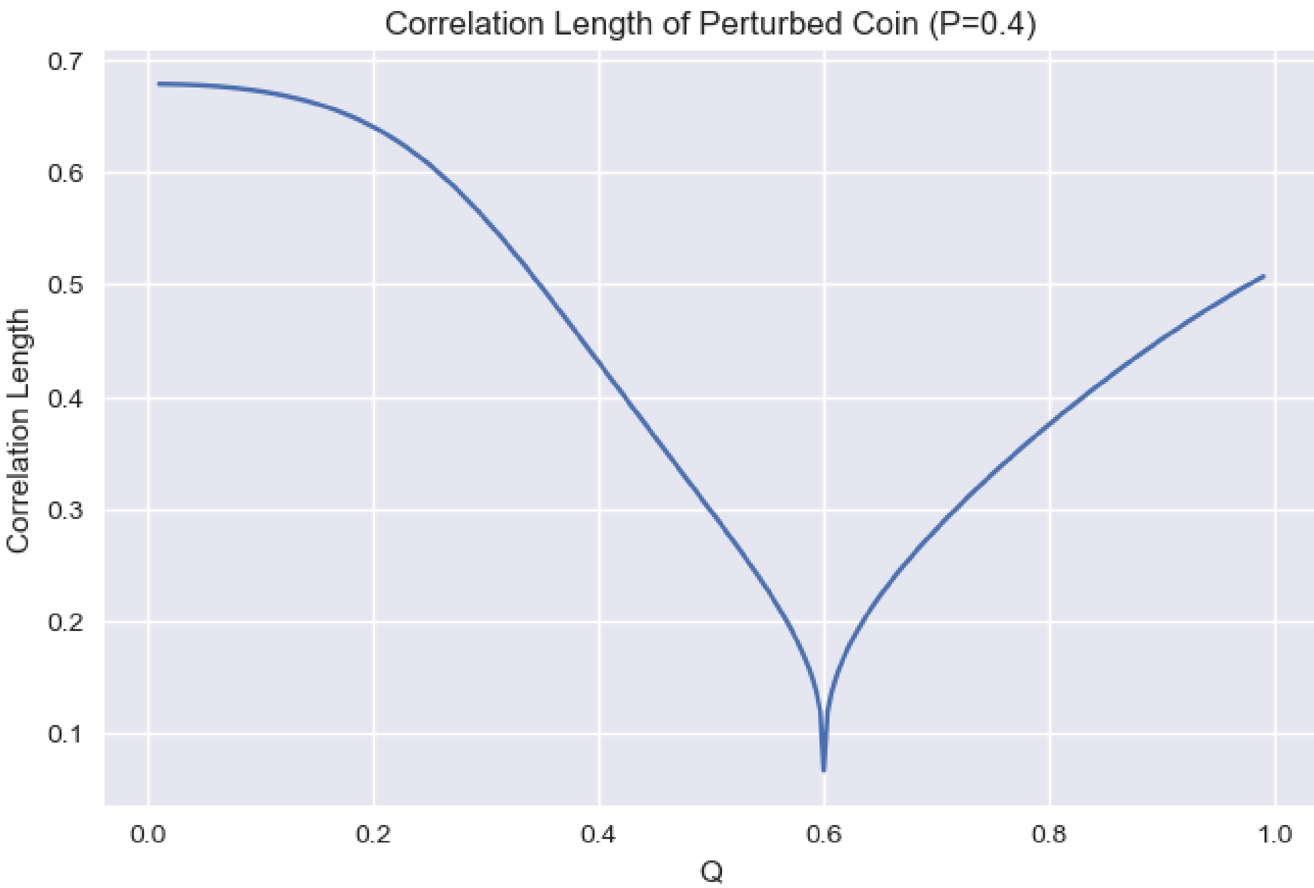
The site matrices are given by the form:

```
[[[0.4 0. ]
  [0.8 0. ]]

 [[0.  0.6]
  [0.  0.2]]]
```

Check Correlation Length

The system becomes memoryless when it approaches the line $p = q$, we can observe this with the plots below.



As we can see, the correlation length drops to 0 when $p = 1 - q$.

Checking the Produced Sequence by the Model

```
[0 1 0 0 0 0 0 0 1 0 0 1 0 1 0 0 0 0 0 1 0 0 0 0 1 0 0 0 0 0 0 0 0 0 0 1 0
 1 0 0]
```

Symmetry Finding for the Perturbed Coin

We now attempt to find the symmetry group generators for the perturbed coin process for a fixed value of $p \neq 1 - q$.

```
The parent hamiltonian in consideration is:
[[ 0.3194  0.075   0.0564 -0.0486 -0.3323 -0.0486 -0.3048 -0.023 ]
 [ 0.075   0.4732 -0.1555 -0.2888  0.0906 -0.2888 -0.0716 -0.1984]
 [ 0.0564 -0.1555  0.0729  0.1007 -0.1216  0.1007 -0.0538  0.0477]
 [-0.0486 -0.2888  0.1007  0.7671 -0.0587 -0.2329  0.0464 -0.1515]
 [-0.3323  0.0906 -0.1216 -0.0587  0.4087 -0.0587  0.3171 -0.0278]
 [-0.0486 -0.2888  0.1007 -0.2329 -0.0587  0.7671  0.0464 -0.1515]
 [-0.3048 -0.0716 -0.0538  0.0464  0.3171  0.0464  0.2908  0.022 ]
 [-0.023  -0.1984  0.0477 -0.1515 -0.0278 -0.1515  0.022   0.9009]]

================================================================================
SYMMETRIC GENERATORS ANALYSIS
================================================================================

⚠️  No symmetry found: The null space is empty.
================================================================================
```

Now for the case $p = q = 0.5$,

```
The parent hamiltonian in consideration is:
[[ 0.5  0.  -0.5  0.   0.   0.   0.   0. ]
 [ 0.   0.5  0.  -0.5  0.   0.   0.   0. ]
 [-0.5  0.   0.5  0.   0.   0.   0.   0. ]
 [ 0.  -0.5  0.   0.5  0.   0.   0.   0. ]
 [ 0.   0.   0.   0.   0.5  0.  -0.5  0. ]
 [ 0.   0.   0.   0.   0.   0.5  0.  -0.5]
 [ 0.   0.   0.   0.  -0.5  0.   0.5  0. ]
 [ 0.   0.   0.   0.   0.  -0.5  0.   0.5]]
Verifying that the MPS is in the ground space of the parent hamiltonian:
ph_3_3 @ state =  [0. 0. 0. 0. 0. 0. 0. 0.]

================================================================================
SYMMETRIC GENERATORS ANALYSIS
================================================================================

Found 1 generator(s) in the null space:
--------------------------------------------------------------------------------

Generator 1:
  Null space basis vector index: 0
  Coefficients in local su(2) basis (orthonormalized): [1. 0. 0.]

  Basis generators in su(2) (with non-zero coefficients):
    [0] (coeff: 1.0):
[[0.+0.j 1.+0.j]
 [1.+0.j 0.+0.j]]

  Local su(2) generator (Hermitian enforced):
[[0.+0.j 1.+0.j]
 [1.+0.j 0.+0.j]]

--------------------------------------------------------------------------------

Orthonormalized null space basis (1 vectors):

Basis vector 1 (norm = 1.0000):
[1. 0. 0.]

================================================================================
```

We see that the hamiltonian is of the form $H = I \otimes (I - \sigma_x) \otimes I$.

```
Checking the form of the Hamiltonian:
[[ 0.5  0.  -0.5 -0.   0.   0.  -0.  -0. ]
 [ 0.   0.5 -0.  -0.5  0.   0.  -0.  -0. ]
 [-0.5 -0.   0.5  0.  -0.  -0.   0.   0. ]
 [-0.  -0.5  0.   0.5 -0.  -0.   0.   0. ]
 [ 0.   0.  -0.  -0.   0.5  0.  -0.5 -0. ]
 [ 0.   0.  -0.  -0.   0.   0.5 -0.  -0.5]
 [-0.  -0.   0.   0.  -0.5 -0.   0.5  0. ]
 [-0.  -0.   0.   0.  -0.  -0.5  0.   0.5]]
```

Finding Virtual Symmetry

Next we find the associated virtual symmetry of the system

```
Physical generator G:
[[0.+0.j 1.+0.j]
 [1.+0.j 0.+0.j]]

Virtual generator found:
  Phase: 1.000000+0.000000j
  Error: 3.055051e-06
  Relative error: 3.055051e-06
  Generator K:
[[ 1.+0.j -0.+0.j]
 [-0.+0.j -1.+0.j]]

The canonical form of the MPS is:
[[[ 0.5  0.5]
  [ 0.   0. ]]

 [[ 0.5 -0.5]
  [-0.   0. ]]]
Verifying that the MPS is in the ground space of the parent hamiltonian:
[0. 0. 0. 0. 0. 0. 0. 0.]
```

Verifying the Generator Equation

Hence, we can conclude that rotations generated by σ_x on the physical level can be projected onto the virtual level as rotations generated by σ_z. As they are both 1-D subspaces of SU(2), the 2-cocycle is trivially constant.

```
==============================================================================
Generator 1
==============================================================================

For angle θ = 0.0 π

Site matrix A[0]

Physical                              |   Virtual
-----------------------------------   |   -----------------------------------
[[0.5+0.j 0.5+0.j]                    |   [[0.5+0.j 0.5+0.j]
[0. +0.j 0. +0.j]]                    |   [0. +0.j 0. +0.j]]

Site matrix A[1]

Physical                              |   Virtual
-----------------------------------   |   -----------------------------------
[[ 0.5+0.j -0.5+0.j]                  |   [[ 0.5+0.j -0.5+0.j]
[-0. +0.j  0. +0.j]]                  |   [-0. +0.j  0. +0.j]]

Relative Phases: 0.0 π
0.0 π
0.0 π
-0.0 π

==============================================================================

For angle θ = 0.25 π

Site matrix A[0]

Physical                              |   Virtual
-----------------------------------   |   -----------------------------------
[[0.3536+0.3536j 0.3536-0.3536j]      |   [[ 0.5+0.j  -0. +0.5j]
[0.    -0.j     0.    +0.j    ]]      |   [-0. -0.j   0. -0.j ]]

Site matrix A[1]

Physical                              |   Virtual
-----------------------------------   |   -----------------------------------
[[ 0.3536+0.3536j -0.3536+0.3536j]    |   [[ 0.5-0.j  -0. -0.5j]
[-0.    +0.j      0.    +0.j    ]]    |   [-0. +0.j   0. +0.j ]]

Relative Phases: 0.25 π
-0.75 π
0.25 π
-0.75 π

==============================================================================

For angle θ = 0.3333 π

Site matrix A[0]

Physical                              |   Virtual
```

```
-----------------------------------   |   -----------------------------------
[[0.25+0.433j 0.25-0.433j]            |   [[ 0.5 +0.j    -0.25+0.433j]
[0.  -0.j    0.  +0.j   ]]            |   [-0.  -0.j     0.  -0.j   ]]

Site matrix A[1]

Physical                              |   Virtual
-----------------------------------   |   -----------------------------------
[[ 0.25+0.433j -0.25+0.433j]          |   [[ 0.5 -0.j     0.25-0.433j]
[-0.  +0.j     0.  +0.j   ]]          |   [-0.  +0.j     0.  +0.j   ]]

Relative Phases: 0.3333 π
1.0 π
0.3333 π
-1.0 π

======================================================================

For angle θ = 0.5 π

Site matrix A[0]

Physical                              |   Virtual
-----------------------------------   |   -----------------------------------
[[0.+0.5j 0.-0.5j]                    |   [[ 0.5+0.j -0.5+0.j]
[0.-0.j  0.+0.j ]]                    |   [-0. -0.j  0. -0.j]]

Site matrix A[1]

Physical                              |   Virtual
-----------------------------------   |   -----------------------------------
[[0.+0.5j 0.+0.5j]                    |   [[0.5-0.j 0.5-0.j]
[0.+0.j  0.+0.j ]]                    |   [0. +0.j 0. +0.j]]

Relative Phases: 0.5 π
0.5 π
0.5 π
0.5 π

======================================================================

For angle θ = 0.75 π

Site matrix A[0]

Physical                              |   Virtual
-----------------------------------   |   -----------------------------------
[[-0.3536+0.3536j -0.3536-0.3536j]    |   [[ 0.5-0.j  -0. -0.5j]
[-0.    -0.j     -0.    +0.j    ]]    |   [-0. +0.j   0. +0.j ]]

Site matrix A[1]

Physical                              |   Virtual
-----------------------------------   |   -----------------------------------
[[-0.3536+0.3536j  0.3536+0.3536j]    |   [[ 0.5+0.j  -0. +0.5j]
[ 0.    +0.j     -0.    +0.j    ]]    |   [-0. -0.j   0. -0.j ]]
```

```
Relative Phases: 0.75 π
-0.25 π
0.75 π
-0.25 π

======================================================================

For angle θ = 1.0 π

Site matrix A[0]

Physical                                |   Virtual
-----------------------------------     |   -----------------------------------
[[-0.5+0.j -0.5-0.j]                    |   [[0.5-0.j 0.5-0.j]
[-0. -0.j -0. +0.j]]                    |   [0. +0.j 0. +0.j]]

Site matrix A[1]

Physical                                |   Virtual
-----------------------------------     |   -----------------------------------
[[-0.5+0.j  0.5+0.j]                    |   [[ 0.5+0.j -0.5+0.j]
[ 0. +0.j -0. +0.j]]                    |   [-0. -0.j  0. -0.j]]

Relative Phases: 1.0 π
-1.0 π
1.0 π
-1.0 π

======================================================================
```